

National College of Ireland

MSCCLOUD1JAN26I and MSCCLOUD1JAN26I_B

Due Date: Monday 20th July 2026

Lecturers: Alejandro Flores Lamas and Vikas Sahni

Cloud Machine Learning Continuous Assessment (CA) Type: Project

Weight: The assignment will be marked out of 100. The weight of this assignment is 50% of the total marks for this module.

Instructions: Work in groups of 3-4 students

SUBMISSION DETAILS:

A. Report (One per team):

1. A report of up to 6 pages that addresses the assignment is required to be submitted online on Moodle before the deadline. There will be a 10% penalty for each extra page.
2. The report must be formatted according to the [IEEE - Manuscript Templates for Conference Proceedings](#), A4 page size, and must be submitted as a MS-Word file.
3. Make sure to acknowledge the original sources for your investigation as appropriate, including the use of AI. You **MUST** include the NCI assignment submission cover page.
4. The use of **grammar tools** such as Grammarly is **PROHIBITED**.
5. The assignment submission coversheet does not count in the 6-page limit.
6. Everything else, such as references, appendices etc. is included in the 6-page limit.
7. Only **ONE** team member, whose first name is alphabetically the first, **MUST** upload it.

B. Source code and dataset(s) used in the project (Individual):

The source code and dataset(s) used in the project must be submitted online on Moodle before the deadline as a single compressed .ZIP file. Also, every student must submit a video of maximum 4 minutes to show the correct execution of their code. There will be a 20% penalty for every 30 seconds over 4 minutes. Videos that are **speeded up will NOT BE GRADED**.

C. ML Strategy and individual feedback for the project (Individual):

Each student must upload a short (maximum one page) document describing the machine learning algorithm implemented by them. This document must also contain a table listing in your opinion what percentage each team member has contributed to the overall project. For example, A – 30%, B – 20%, C – 25% and D – 25%.

D. Presentation (Team):

1. **The presentation/demo will be held in class** after the submission deadline.
2. If you exceed the maximum time limit of 9 minutes, a 20% deduction per additional 30 seconds will be applied.

3. **All team members MUST be present.** Those who are absent during the presentation will be marked absent and will be awarded ZERO marks.

TURNITIN: All report submissions will be electronically screened for evidence of academic misconduct (i.e., plagiarism and collusion)

Project Brief

1. Introduction

This project is aimed at achieving the module Learning Outcomes listed below:

LO2	Clean and transform datasets in preparation for data mining, and build and evaluate machine learning models to extract knowledge from various datasets.
LO4	Evaluate and utilise cloud computing technologies and services for data collection, storage and mining when designing and implementing data driven applications.

2. Project Description

Students are required to propose and execute a research project using machine learning **techniques in the scope of the course as a team**. The project should relate to a real-world problem in any topic. Students should make use of cloud computing technologies and services for **data collection, storage and mining**. While doing the project, students should consider the **ethical aspects** regarding the datasets and machine learning algorithms used.

The overarching focus of the project is to **develop a cloud-based solution of a problem with machine learning methods/techniques**. Your solution should aim at roughly balancing the three paradigms, namely performance, scalability and latency and throughput, of your models. The development pipeline should include the following aspects: data collection process in cloud, transformation strategies applied (locally or in cloud). In order to view data characteristics, you should apply statistical analysis tools. **Each team member will apply a different ML technique and build their own model**. Each method should be applied to answer the **same specific research question(s) aligned to the overall goal(s)** of the project. It is also expected that the application of each method is accompanied by an appropriately sized review **documenting** pertinent and contemporary approaches in the literature that can both inform the application of a method as well as **justify its potential merit(s)**. You must use **appropriate evaluation techniques and tools** to measure the performance (in terms of quality and latency) of your methods / models. You should **perform** a comprehensive manual error analysis on the outputs produced by your models and baseline models which have been considered for your investigation. Finally, **the models built by each team member should be compared** using standard metrics to show the efficiency of the strategy and its implementation in a cloud environment.

Projects will be assessed based on their **novelty, technical quality, insightfulness, depth, clarity, robustness, quality of writing and reproducibility**. Code and datasets (different for each team) must also be submitted with the report. Algorithms and resources used in a report should be described as completely as possible to allow reproducibility. This includes experimental methodology, empirical evaluations, and results. The reproducibility factor will play an important role in the assessment of each submission.

It is essential that the project shows unambiguous evidence of:

1. A critical analysis of fundamental machine learning methodologies to assess best practice guidance when applied to computational problems (e.g. data mining, regression, classification, etc.) in the specific context of the project.
2. The extraction, combination, transformation, exploration, and cleaning of datasets in preparation for building machine learning methods used in the project.
3. Making use of cloud computing technologies and services for data collection, storage and mining.
4. The building and evaluation of machine learning models in the scope of the course on a variety of datasets and parameters; producing the learning curve of your ML models on varying sized datasets, data domains and number of parameters.
5. Reporting performance (quality, latency and throughput) and scalability of the machine learning models in varying-sized datasets. The discussion of performance should be orientated around multiple notions of performance and across a variety of parameters (e.g. latency).
6. The extraction, interpretation, and evaluation of information and knowledge that is drawn from the datasets used.
7. The critical review of relevant machine learning research to afford the assessment of research methods applied in the project.
8. A prototype showing facilitating services (e.g. real-time or batch) with your models via cloud to the end-users.

3. Report Guidelines

The final report must follow the IEEE conference format and should be between 5 to 6 pages in length. The NCI assignment submission coversheet pages are not included in the count of pages.

For this report IEEE referencing style should be used.

The following structure is suggested for the report:

Abstract: 100–150 words providing a high-level description of the project, its core findings, and the domain of the datasets (not necessarily in this order).

Introduction: Remainder of 1st page. It should motivate the work, present and discuss the research question(s) / objective(s) of the project and (optionally) provide a concise overview of the following sections (max 1–2 lines per each).

Related Work: Half page — this should not only summarize the related works, but also critically evaluate their key positive and negative aspects with respect to the topic and domain of the project, i.e., how well/badly do the related works address your question(s) / objective(s), what aspects are useful to consider, what are the limitations, etc. Also include here a discussion on the previous uses of the datasets and the methods applied. If you plan to reuse a method already applied to this dataset, discuss what you expect to gain by doing this.

Methodology: This section can be named differently. But it should describe how you have approached answering your question. Additional (technical) details can also be discussed here. Essentially, you should recount how you applied ML models in the cloud to facilitate your research question(s). You should also include here a discussion on key preliminary aspects of the methodology, such as how the datasets have been prepared for study (i.e., the pre-processing, and transformation stages).

Implementation: This section should have a separate subsection about the individual work of each team member.

Evaluation: How have you used your methodology to answer the research question(s) (evaluation methodology)? How do you know that a method is good? What performance measures have you selected and why (discuss how the choice of performance measures is appropriate)? If you have to parametrize part of an approach, how have you done that, and why were these choices made, and what impacts can different parameters have on your results? You should also discuss the results in detail in this section: what are their implications? What do they show / not show? etc. A discussion on sampling methods is expected here too. This section should include a subsection 'error analysis' which will present a thorough and comprehensive error analysis on the outputs of your systems / models including the baselines.

Conclusions and Future Work: Summarize your findings and discuss limitations / extensions you would do next to improve or extend your study if you had more time. Summarize the (partial) answer to the research question(s) at a high level, and note the key implications of your findings with respect to the methods studied.

Contributions: A table should be used to show individual contributions in the project. The table should clearly highlight who carried out what part of the project (e.g. design, development, experiments and report preparation).

References: Include a list of all references used in your report.

4. Grading Rubric

This project will be graded using the marking grid shown in Table 1. Note that marks in Table 1 represent percentage. THE FINAL MARK MUST BE 40% OR ABOVE TO ACHIEVE A PASS.

NOTE: All team members MUST be present for the presentation/demo to be held in class. Those who are absent during the presentation will be marked absent and will be awarded ZERO marks.

Extension/Re-run

Should any student miss the assessment deadline with a valid reason, he/she can now apply for an application for coursework Extension/Re-run Form online, via NCI360.

N.B.

All submissions will be electronically screened for evidence of academic misconduct (plagiarism and collusion).

Table 1. Grading Rubric

	0 – 39%	40 – 49%	50 – 59%	60 – 69%	70% +	Marks
PROJECT SPECIFICATION 10% Project Objectives: definition & attainment	Objectives are inadequately specified	Objectives may be specified but not achieved.	Objectives have been clearly specified and achieved to some extent.	Objectives have been clearly specified and are appropriate. The objectives have been fully achieved.	Objectives have been clearly specified and are creative and appropriate. Objectives have been fully achieved or surpassed	0-10
LITERATURE REVIEW 10% Use of and critical understanding of relevant theory	Very limited use of theory and concepts	Limited use of theory and concepts	Reasonable use of theory and concepts but lack of breadth in literature reviewed.	Comprehensive use of relevant theory and concepts. Evidence of breadth in literature review.	Critical application and critique of relevant theory and concepts. Evidence of breadth and depth of literature reviewed.	0-10
ARTEFACT / PRODUCT DEVELOPMENT 30% Relevance and justification of methodology utilised and the ability to carry out the specified methodology	Alternative methodologies not considered and method selected not justified. Significant errors in the application of methodology.	Alternative methodologies have not been considered and only limited justification for selected methods. Some errors in application.	Alternative methodologies have not been fully considered, though a reasonable attempt made at justifying its use. The methodology has been completely carried out.	Alternative methodologies have been considered and the methodology selected has been justified. The methodology has been competently carried out.	Alternative methodologies have been fully considered and the chosen methodology fully justified. The application has been rigorously carried out.	0-30
EVALUATION 25% Analysis of the research findings and ability to understand the study's implications and limitations	Pedestrian or poor attempt to analyse the findings. Implications and limitations not understood.	Some attempt at analysis although with some problems. Implications and limitations not well understood.	A reasonable attempt to analyse the findings. However, the synthesis of data collected, and theory is not fully achieved. Implications not fully appreciated.	Rigorous analysis of findings. Demonstrates the ability to synthesise data collected and relevant theory. Shows an understanding of implications of the study.	Rigorous and creative analysis of findings. Demonstrated the ability to synthesise data collected and relevant theory. Insightful conclusions which appreciate limitations and implications of the study.	0-25
DOCUMENT STRUCTURE, PRESENTATION & REFERENCING 15% Quality of document presentation – including referencing, structure, grammar, and spelling.	Poor presentation with inconsistent referencing. Inadequate grammar and spelling.	Reasonable presentation but inconsistent in referencing. Acceptable grammar and spelling.	Satisfactory presentation with consistent referencing and clear structure. Acceptable grammar and spelling.	Good presentation and structure with rigorous referencing. Correct grammar and spelling	Excellent presentation and structure with rigorous referencing. Correct grammar and spelling.	0-15
VIVA 10% Quality of Oral Presentation – including handling of questions from examiners	Unacceptable oral presentation and handling of questions.	Poor oral presentation and weak handling of questions.	Neat oral presentation and acceptable handling of questions.	Clear presentation with good handling of questions.	Excellent well directed presentation with impeccable handling of questions.	0-10